# Adaptive Staircase Experiment
# PSY310 Lab in Psychology
# Lab Report

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**GitHub Link:** **https://github.com/saiman-ui/Lab-Psychology/tree/Adaptive-Staircase**

# Introduction

Because they offer an effective method of estimating sensory thresholds while reducing needless trials, adaptive staircase techniques are frequently employed in psychophysics. Depending on the participant's performance, the technique—also known as the up-down method or the method of limits—dynamically modifies the stimulus intensity. The utility of staircase designs in sensory research was shown by early studies (Dixon & Mood, 1948; Cornsweet, 1962), and their dependability for threshold estimate was codified in later work (Levitt, 1971; Leek, 2001). These days, adaptive stairs are frequently used in fields including visual perception (Kingdom & Prins, 2010), auditory assessment (Treutwein, 1995), and even taste and smell discrimination (Kaernbach, 1991). In many experimental circumstances, they outperform fixed-step psychophysical approaches due to their efficiency and lower computer cost.

# Method

*Participant*

The experiment was finished by one participant, a 20-year-old female Ahmedabad University undergraduate. Before the study began, she gave her informed consent. She had to determine the tilt direction of a grating stimulus as part of the test.

*Materials and Procedure*

PsychoPy (v2024.2.2) was used to program the experiment, which was carried out on a desktop computer with a 1920 x 1080 px screen. Each trial started with a 100 ms fixation cross and progressed to a 500 ms sinusoidal grating with a Gaussian mask. On each trial, the tilt orientation was given at random to either the left or the right.

The range of stimulus tilt values was 1° (minimum) to 10° (highest). The stairway had step sizes of 2, 1, 3, and 0.5 and was designed according to the one-up/three-down rule. The staircase automatically adjusted intensity based on response accuracy over the course of 80 trials. The keyboard was used by participants to respond: the left arrow indicated a left tilt, while the right arrow indicated a right tilt.

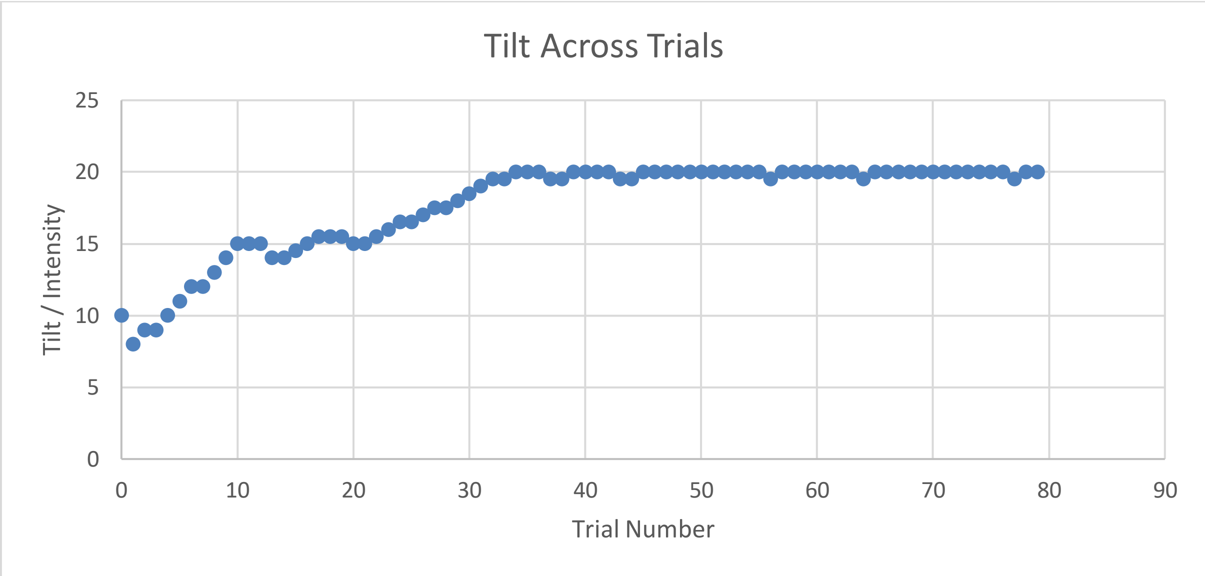


*Figure 1. The adaptive staircase varied tilt intensity across 80 trials, gradually converging toward the participant's threshold around 17.5°.*

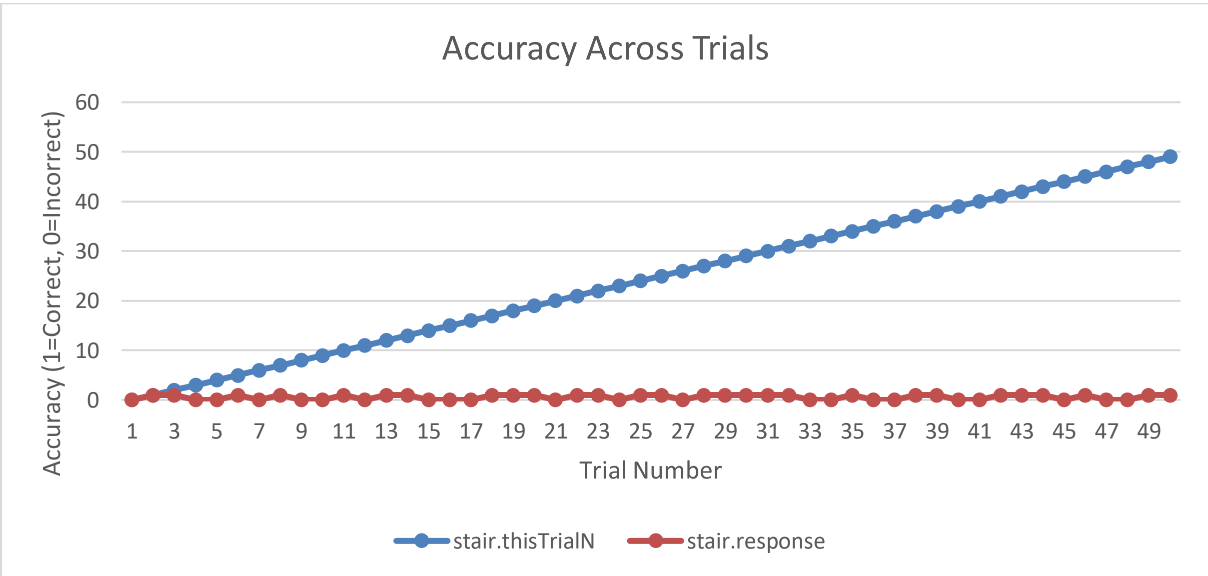


*Figure 2. Accuracy fluctuated between correct (1) and incorrect (0) responses across trials, with an overall performance of about 68%.*

# Results

The staircase procedure identified three reversal points at angles of 12.0°, 19.5°, and 19.5°. Calculating the average of these values resulted in a difference threshold estimate of 17.0°. Although this is based on fewer than the suggested five to ten reversals, it still offers an estimation of the participant's sensitivity. Figures 2 and 3 demonstrate the variations in tilt intensity and response accuracy over the course of the 80 trials.

## Discussion

The evaluation of the responses from the participant indicated a threshold value of 17.0°. This implies that the participant needed considerably larger tilt variations to consistently notice changes in orientation. In contrast to typical results found in psychophysical studies, where trained observers frequently report thresholds below 5° (Levitt, 1971; Kingdom & Prins, 2010), the current estimate is comparatively high. The limited number of reversals (only three) diminishes the reliability, and this finding should be approached with care.

Nonetheless, the adaptive staircase proved to be effective in modifying stimulus intensity in real-time. This methodology is still useful for estimating perceptual thresholds with fewer trials compared to fixed-step techniques (Leek, 2001; Treutwein, 1995). Future studies involving more trials and a larger participant pool would yield a more consistent estimate and enhance external validity.

# References

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